

Name: Team Timbuktu

Date: 2/11/25

Element E: Application of STEM principles and practices

All elements of Stem were incorporated into the design of the crane. Through chaotic, yet extensive planning, the crane was designed with the goal of ease of use in mind. Before anything was built, multiple rough sketches, 2 full 3D models, and several parts/subsystems were 3D modeled as well.

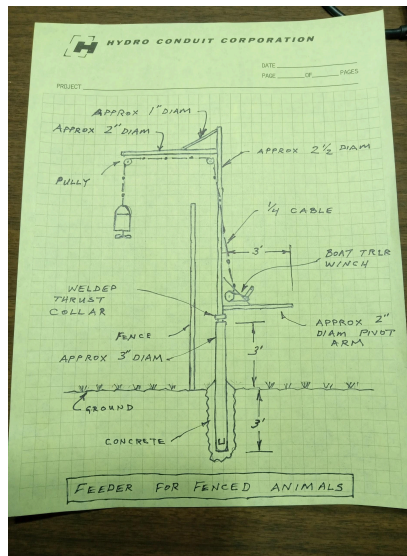
Material science was used in several areas: making sure the wood structure could take the forces, making sure the nylon cord could hold the weight of the toy, and making sure the pulleys for said cord could also hold the weight. Engineering concepts, principles, and processes were used to ensure that the crane was cheap (as we had to work within a budget), effective, user-friendly, and repairable. Cost was valued above the others and as such, changed our initial design to use things like 4x4s instead of Unistrut, and hand pulling the rope instead of a crank/winch.

An engineering major came by the class and reviewed the designs of the groups, and he largely agreed with the idea we had. He did say we should see if there was a cheaper way to make the base structure, which was eventually found (4x4s), and gave us some tips on how to cut the Unistrut.

The grandfather (a civil engineer) of one of the group members was asked if our design was viable, and drew a sketch of what he would build for the task. That sketch was used as proof that we were going in the right direction and helped us continue the way we were going, and convinced us to do away with part of the design and replace it by burying the bottom of the main post and surrounding it in concrete.

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The handlers for the Elands joined a Zoom call and gave us parameters that we had to follow, and even more when we went to the in-person meeting. The only thing this influenced was the size of the structure; the rest we already had figured out by then. Throughout the design process, one of the group members' parents helped with the design as well. The mother of that member has done construction management for the past decade or more, and was able to point out deficiencies, errors, points of stress, and ways to improve. Because of her advice, the idea to have 8" lag bolts screw the horizontal to the vertical was removed, the diagonal 2x4 was replaced with steel cable and a turnbuckle attached to some eye hooks, the attachment point for said steel cable was lowered, the L brackets on the back of the vertical post were removed, the L bracket at the side of the joint was replaced with a T-bracket, rebar was added through the bottom of the vertical board, and the height of the vertical post was increased from 12 feet to 16.

The 8" lag bolts were removed because they would only serve to weaken the top of the vertical post because the amount of material removed could cause the wood to split in the direction the torque is applied to the lag bolt, and if the rest of the structure is built

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correctly, there shouldn't be stress applied to the bolt anyway so it would be useless. The diagonal 2x4 did serve a good purpose, but the method of attachment was a diagonal pocket hole using a lag bolt- far more difficult than using an eye hook to hold steel cable. The L-brackets on the back of the vertical post weren't serving a purpose as the steel cable was taking the torque instead of the brackets, which likely would have bent under the load. The side L-brackets were also a good idea; however, T-brackets do the same thing, but better. The rebar at the bottom of the vertical post was added to more properly bind the concrete to the wood and allow a stronger base. The height of the post was increased to allow the majority of toys to pass over the fence and be more easily replaced.



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After some of the changes in question:

